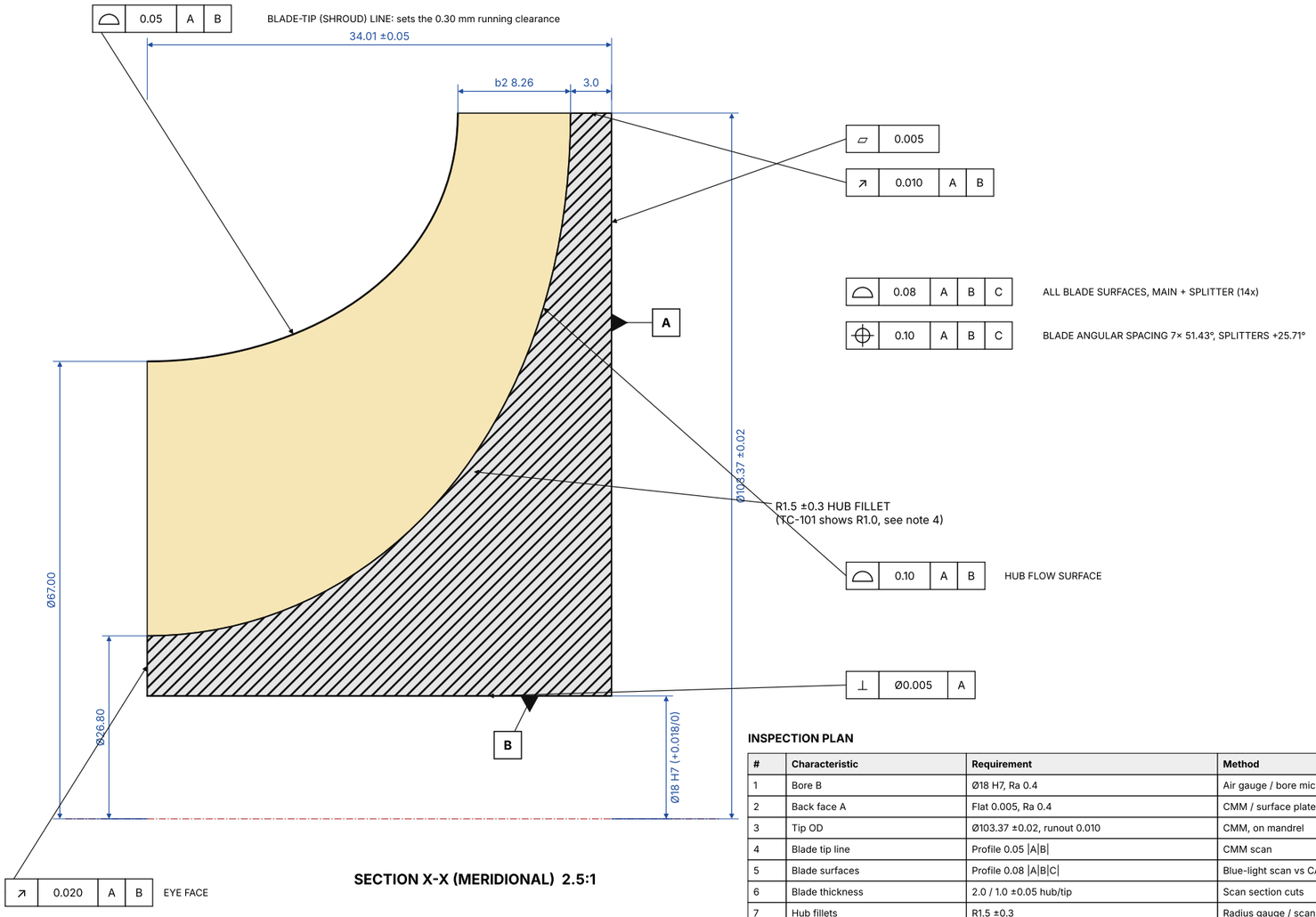
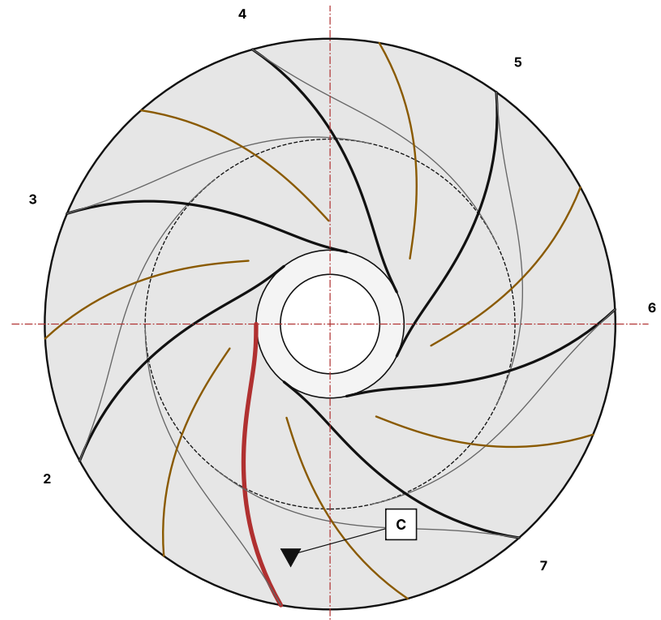


SECTION X-X (MERIDIONAL) 2.5:1



VIEW FROM INLET 0.9:1 (BLADE 1 = DATUM C)

ROTATION CW VIEWED FROM INLET. Red = blade 1 (datum C), brown = splitters

INSPECTION PLAN

#	Characteristic	Requirement	Method
1	Bore B	Ø18 H7, Ra 0.4	Air gauge / bore mic
2	Back face A	Flat 0.005, Ra 0.4	CMM / surface plate
3	Tip OD	Ø103.37 ±0.02, runout 0.010	CMM, on mandrel
4	Blade tip line	Profile 0.05 |A|B|	CMM scan
5	Blade surfaces	Profile 0.08 |A|B|C|	Blue-light scan vs CAD
6	Blade thickness	2.0 / 1.0 ±0.05 hub/tip	Scan section cuts
7	Hub fillets	R1.5 ±0.3	Radius gauge / scan
8	Flow-surface finish	Ra ≤ 0.8 µm	Profilometer, 3 blades
9	Balance	G1.0: U ≤ 0.073 g·mm	2-plane balancer
10	Overspeed	34,500 rpm (115 %), growth ≤ 0.01	Spin pit, re-measure 1, 3

KEY DATA

Item	Value
Finished mass	231 g
Blades	7 main + 7 splitter, 40° backswept
Tip / exit width	Ø103.37 / b2 8.26
Max blade height	20.1 mm at LE
Tightest passage	5.43 mm at m = 0.32
Design speed	30,000 rpm, U2 162 m/s
CAM plan	MFG-110, MFG-120
Blade data	cad_curves/stage1/

NOTES

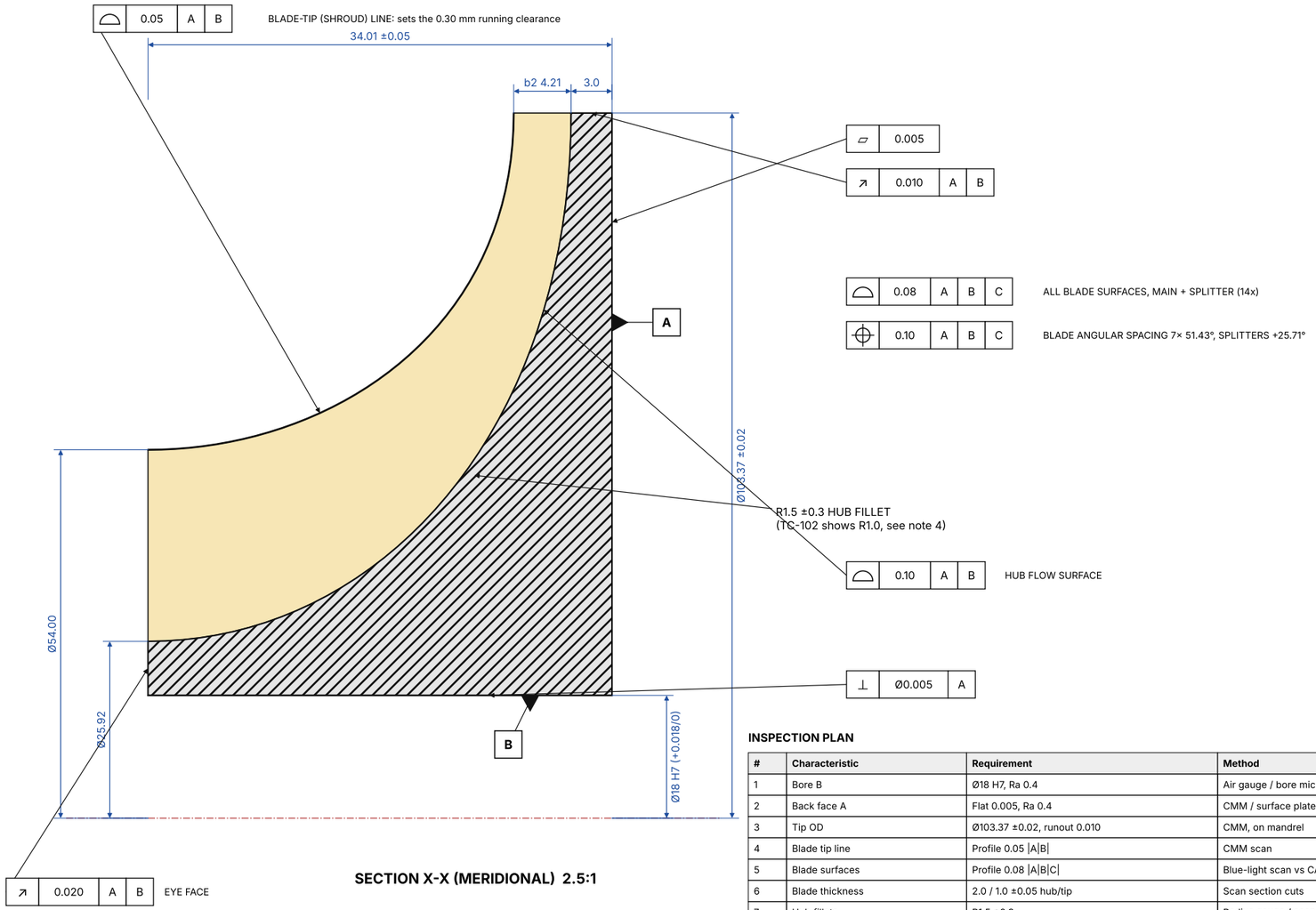
1. All dimensions after machining, at 20 °C. General tolerances ISO 2768-mK.
2. Datums: A = back face, B = bore (A and B finished in one lathe setup, OP30), C = blade 1.
3. Flow surfaces (hub, blades) Ra ≤ 0.8 µm, no visible step between toolpaths > 0.01 mm.
4. Hub fillets R1.5 replace the R1.0 on TC-10x so the Ø3 finishing ball cuts them (docs/05).
5. Leading edges elliptical 2:1, hand-blended after OP40. TE as machined. Other edges 0.2 max.
6. Balance ISO 21940-11 G1.0 in two planes (eye nose, back rim); remove metal only at rim.
7. Spin test at 115 % speed before final balance; re-check items 1 and 3.
8. Mark S/N and a blade-1 dot on the back face, r < 25 mm, low-stress dot peen.
9. Study drawing for a reverse-engineered model - not for production without stress review.

Reverse-engineered study model - NOT OEM geometry. Third-angle projection. General tolerance +/-0.5 mm unless noted.

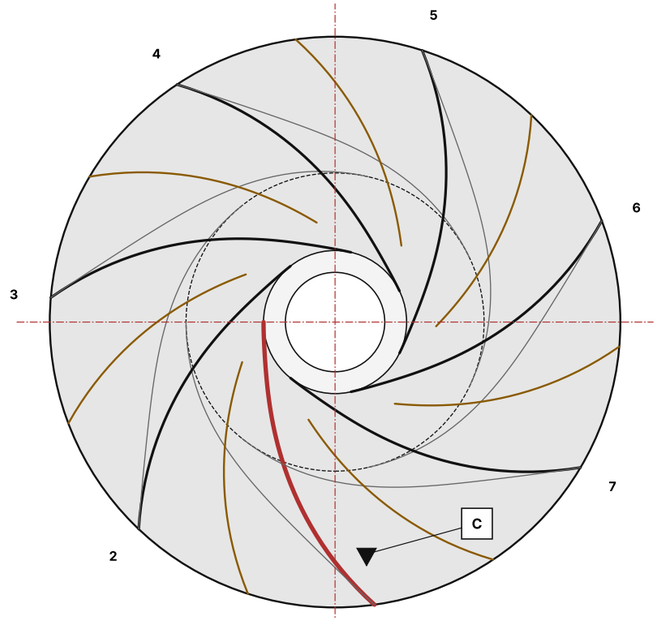
MATERIAL: 7075-T651 bar (UT inspected) or 2618-T61 forging

IMPELLER STAGE 1 - MACHINING

Turbocor-class 2-stage maglev centrifugal compressor (R-513A)	SHEET 1 OF 4
DWG MFG-101 SCALE 2.5:1 / 0.9:1 UNITS mm	REV A 2026-10-27



SECTION X-X (MERIDIONAL) 2.5:1



VIEW FROM INLET 0.9:1 (BLADE 1 = DATUM C)

ROTATION CW VIEWED FROM INLET. Red = blade 1 (datum C), brown = splitters

INSPECTION PLAN

#	Characteristic	Requirement	Method
1	Bore B	Ø18 H7, Ra 0.4	Air gauge / bore mic
2	Back face A	Flat 0.005, Ra 0.4	CMM / surface plate
3	Tip OD	Ø103.37 ±0.02, runout 0.010	CMM, on mandrel
4	Blade tip line	Profile 0.05 |A|B|	CMM scan
5	Blade surfaces	Profile 0.08 |A|B|C|	Blue-light scan vs CAD
6	Blade thickness	2.0 / 1.0 ±0.05 hub/tip	Scan section cuts
7	Hub fillets	R1.5 ±0.3	Radius gauge / scan
8	Flow-surface finish	Ra ≤ 0.8 µm	Profilometer, 3 blades
9	Balance	G1.0: U ≤ 0.068 g·mm	2-plane balancer
10	Overspeed	34,500 rpm (115 %), growth ≤ 0.01	Spin pit, re-measure 1, 3

KEY DATA

Item	Value
Finished mass	214 g
Blades	7 main + 7 splitter, 40° backswept
Tip / exit width	Ø103.37 / b2 4.21
Max blade height	14.0 mm at LE
Tightest passage	4.84 mm at m = 0.32
Design speed	30,000 rpm, U2 162 m/s
CAM plan	MFG-110, MFG-120
Blade data	cad_curves/stage2/

NOTES

1. All dimensions after machining, at 20 °C. General tolerances ISO 2768-mK.
2. Datums: A = back face, B = bore (A and B finished in one lathe setup, OP30), C = blade 1.
3. Flow surfaces (hub, blades) Ra ≤ 0.8 µm, no visible step between toolpaths > 0.01 mm.
4. Hub fillets R1.5 replace the R1.0 on TC-10x so the Ø3 finishing ball cuts them (docs/05).
5. Leading edges elliptical 2:1, hand-blended after OP40. TE as machined. Other edges 0.2 max.
6. Balance ISO 21940-11 G1.0 in two planes (eye nose, back rim); remove metal only at rim.
7. Spin test at 115 % speed before final balance; re-check items 1 and 3.
8. Mark S/N and a blade-1 dot on the back face, r < 25 mm, low-stress dot peen.
9. Study drawing for a reverse-engineered model - not for production without stress review.

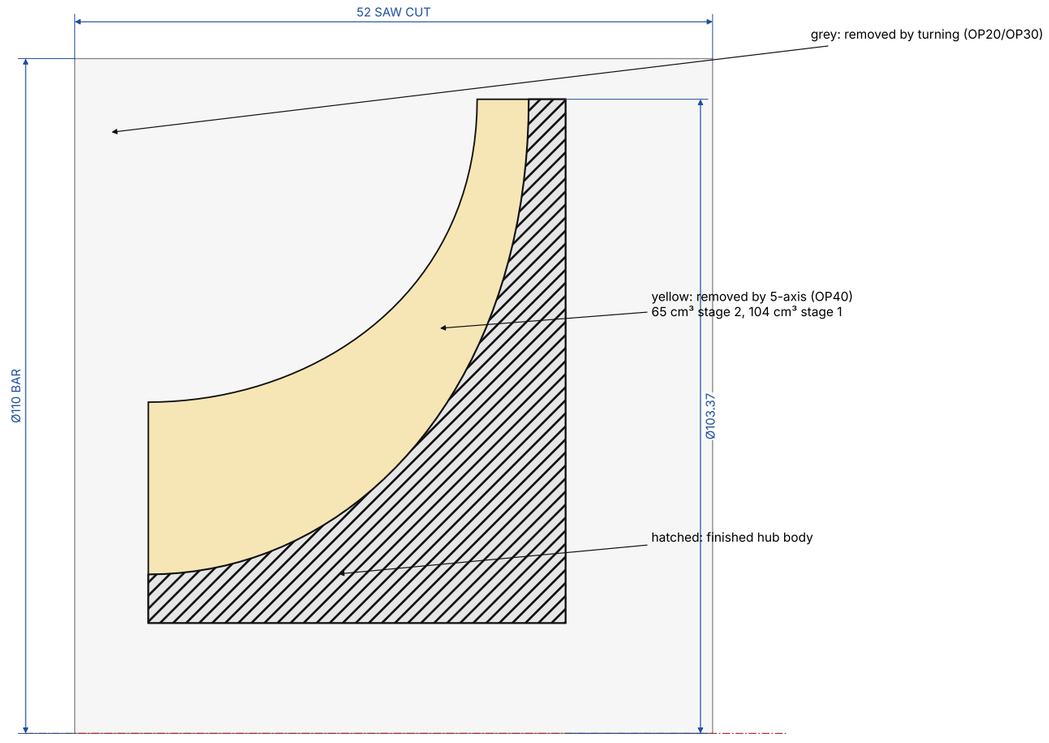
MATERIAL: 7075-T651 bar (UT inspected) or 2618-T61 forging

IMPELLER STAGE 2 - MACHINING

Turbocor-class 2-stage maglev centrifugal compressor (R-513A) | SHEET 2 OF 4

DWG MFG-102 SCALE 2.5:1 / 0.9:1 UNITS mm | REV A 2026-10-27

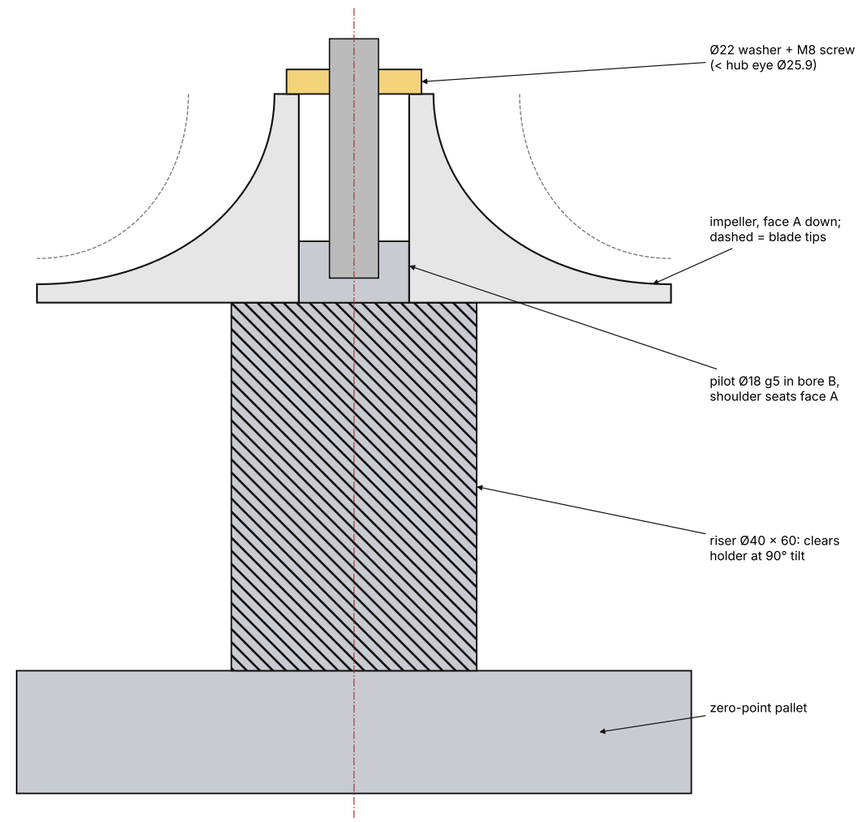
Reverse-engineered study model - NOT OEM geometry. Third-angle projection. General tolerance +/-0.5 mm unless noted.



BLANK SECTION, STAGE 2 SHOWN 2:1

ROUTING

OP	Machine	Work	Datum / holding
10	Saw	Cut Ø110 bar to 52, UT-inspected stock	-
20	CNC lathe	Face eye side, turn blade-tip (shroud) envelope +0.1,	3-jaw on bar OD
		tip OD +0.2, drill + rough bore Ø17	
30	CNC lathe	Face back face A, finish bore B Ø18 H7,	Soft jaws on tip OD
		finish tip OD + tip envelope (one setup with A, B)	
40	5-axis mill	Blades + hub from solid, see MFG-120	Mandrel on A + B
50	Bench	Deburr, blend LEs 2:1, clean	-
60	CMM / scan	Inspection plan items 1-8	A, B, C
70	Balancer	2-plane balance G1.0	Balancing arbor on B
80	Spin pit	115 % overspeed, re-inspect, final balance	Arbor on B



5-AXIS FIXTURE (SECTION) 1:1

NOTES

1. OP30 finishes A, B, tip OD and the tip envelope in one chucking; tip runout and the 0.30 mm clearance depend on it.
2. Tip contour is turned, not milled, so the 5-axis paths never touch the blade-tip line.
3. OP40 creates datum C (blade 1). No dowel needed: probe tip OD + face A, start at 0°.
4. Riser height and washer size set by tool-holder clearance; check in CAM simulation.
5. Same blank and fixture for both stages; stage 1 removes more (longer, taller blades).

Reverse-engineered study model - NOT OEM geometry. Third-angle projection. General tolerance +/-0.5 mm unless noted.

MATERIAL: Bar Ø110 × 52, 7075-T651 (both stages)

IMPELLER BLANK, TURNING & 5-AXIS FIXTURE

Turbocor-class 2-stage maglev centrifugal compressor (R-513A)	SHEET 3 OF 4
DWG MFG-110 SCALE 2:1 / 1:1 UNITS mm	REV A 2026-10-27

OPERATIONS (estimated minutes incl. 1.35x motion factor)

Step	Tool	Strategy	Stock	S2 min	S1 min
OP40-1	T1	5-axis adaptive / multiblade rough	0.30	17	28
OP40-2	T2	Rest rough at splitter LE, corners	0.30	48	71
OP40-3	T3	Blade + hub semi, 0.49 stepover	0.10	81	104
OP40-4	T3	Hub floor, flowline, 0.22 stepover	0	69	69
OP40-5	T3	Blades point-milled, 0.22 stepover	0	111	163
OP40-6	T4	Only if R1.0 fillets are kept	0	12	12
OP40-7	T5	Edge break, not LEs	-	2	2
TOTAL		R1.5 fillets, point-finished blades		327	437
ALT	T3	Stage 2: flank (swarf) finish blades		224	n/a

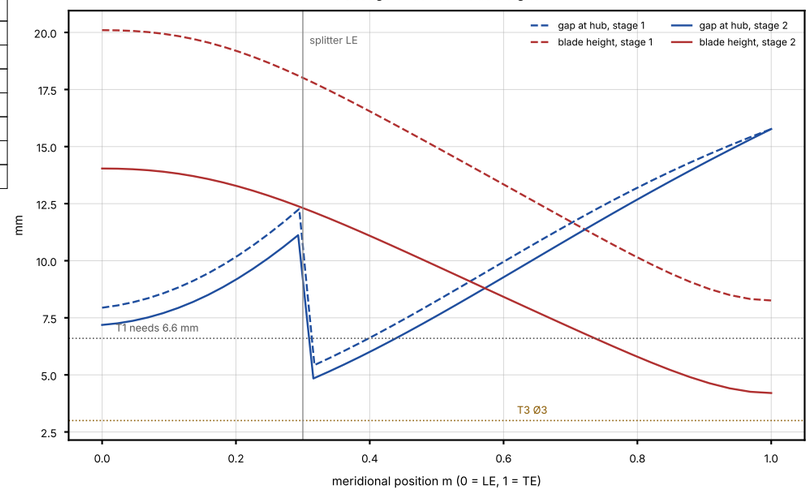
TOOLS (carbide, 15,000 rpm)

T	Tool	F mm/min
T1	Ø5 bull nose R0.5, 3 fl, 20 mm reach	1800
T2	Ø3 bull nose R0.25, 3 fl, 20 mm reach	900
T3	Ø3 tapered ball R1.5, 2° per side, 2 fl	750
T4	Ø2 tapered ball R1.0, 3° per side, 2 fl	450
T5	Ø6 chamfer / deburr, 90°	1800

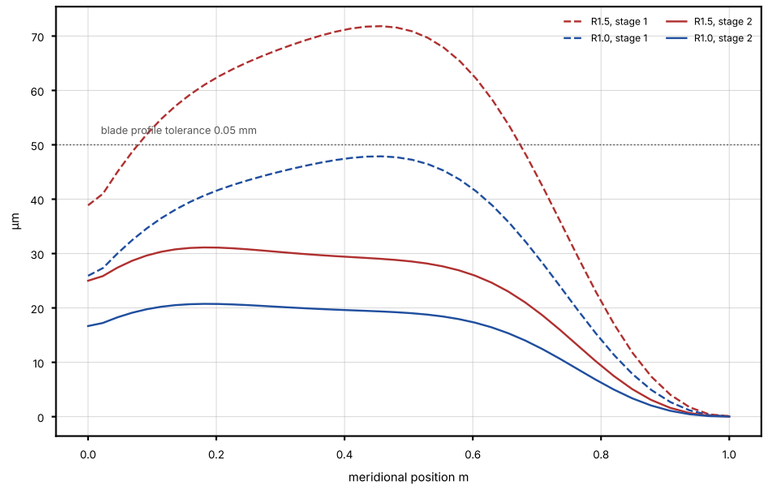
TOOL ACCESS & FLANK-MILLING CHECK

Check	Stage 1	Stage 2
Tightest gap (hub)	5.43 mm	4.84 mm
Blade height there	17.8 mm	12.1 mm
Reach / Ø for T3	5.9	4.0
Max ruling twist	35.6°	23.4°
Flank error, R1.5	0.072 mm	0.031 mm
Flank finish OK (≤0.05)?	no: point-mill	yes
Metal removed (OP40)	104 cm³	65 cm³

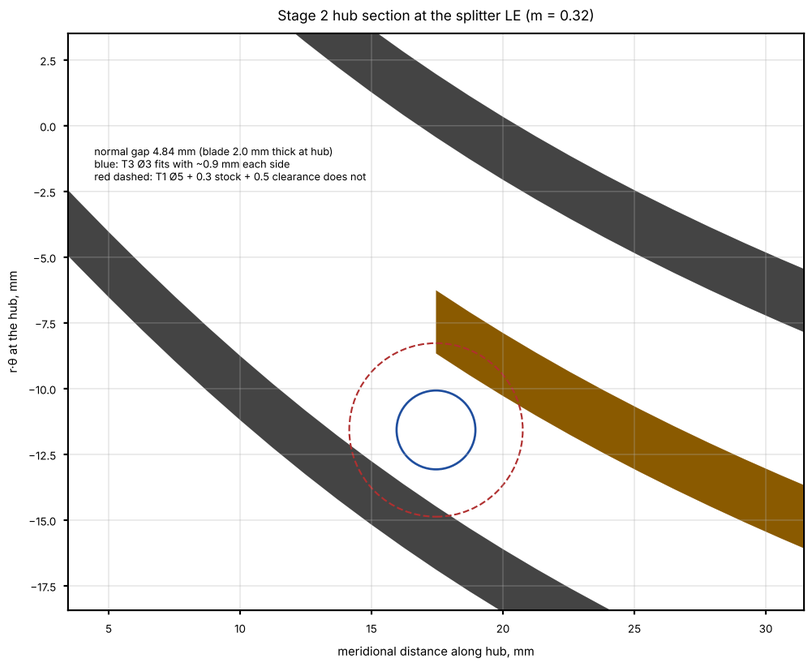
Passage width vs blade height



Flank-milling deviation from ruling twist



Stage 2 hub section at the splitter LE (m = 0.32)



NOTES

1. Blade CAM: hyperMILL Impeller, Mastercam Blade Expert, NX Turbomachinery, SolidCAM Multiblade.
2. Fusion 360 (Manufacturing Ext.): no blade wizard; use swarf + multi-axis contour + rest roughing.
3. Export STEP from SolidWorks; select hub, blade-tip, blade, splitter and fillet faces in the wizard.
4. Tilt strategy: lead 3-5°, lean from blade, interpolate tool axis between hub and shroud.
5. Rough with constant stock; finish hub before blades so the blade root sees the final floor.
6. Finish blades top-down in 2-3 depth steps so uncut metal below supports the thin tip.
7. Simulate full machine + holder; check the Ø3 neck against the next blade at m ≈ 0.3.
8. Times are first-order estimates (scripts/cam_analysis.py); trust the CAM simulation.

Reverse-engineered study model - NOT OEM geometry. Third-angle projection. General tolerance +/-0.5 mm unless noted.

5-AXIS CAM PLAN (OP40)

Turbocor-class 2-stage maglev centrifugal compressor (R-513A)	SHEET 4 OF 4
DWG MFG-120 SCALE NTS UNITS mm	REV A 2026-10-27